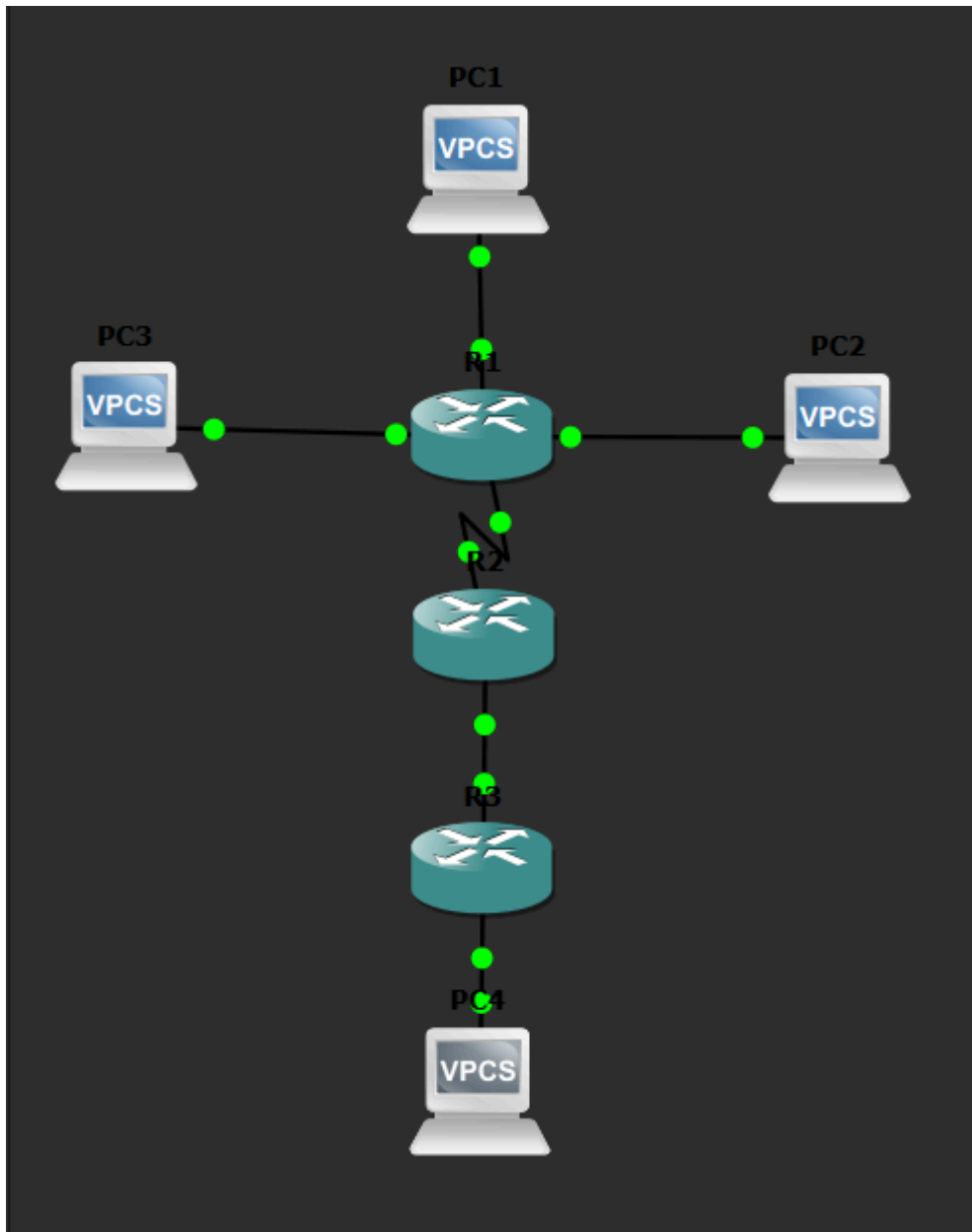


Ejercicio 1 - Estático



#1 Estático

~~10.19.25.0/24~~

10.25.36.0/24

Área	Red	máscara	primer host	último host	broadcast	salto	máscara
lan PC1	10.25.36.0	255.255.255.192	10.25.36.1	10.25.36.62	10.25.36.63	64	$n=6$ $32-6=26$
lan PC2	.64	.224	.65	.94	.95	32	$n=5$ $32-5=27$
lan PC3	.96	.240	.97	.110	.111	16	$n=4$ $32-4=28$
lan PC4	.112	.248	.113	.118	.119	8	$n=3$ $32-3=29$
R1-R2	.120	.252	.121	.122	.123	4	$n=2$ /30
R2-R3	.124	.252	.125	.126	.127	4	$n=2$ /30

tablas de IP

	interfaz	IP	máscara	gateway
lan PC1	R1 Fa0/0	.1	.192	—
	PC1 NCI	.2	.192	.1
lan PC2	R1 Fa0/1	.65	.224	—
	PC2 NCI	.66	.224	.65
lan PC3	R1 Fa1/0	.97	.240	—
	PC3 NCI	.98	.240	.97
lan PC4	R3 Fa0/0	.113	.248	—
	PC4 NCI	.114	.248	.113
R1-R2	R1 S0/0	.121	.252	—
	R2 S0/0	.122	.252	—
R2-R3	R2 Fa0/0	.125	.252	—
	R3 Fa0/1	.126	.252	—

R1

```
R1#
*Mar 1 00:06:22.587: %SYS-5-CONFIG_I: Configured from console by console
R1#write memory
Building configuration...
[OK]
R1#
*Mar 1 00:08:54.259: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to up
R1#
R1#
R1#
R1#show ip interface brief
Interface                IP-Address      OK? Method Status          Protocol
FastEthernet0/0          10.25.36.1      YES manual up              up
Serial0/0                 10.25.36.121    YES manual up              up
FastEthernet0/1          10.25.36.65     YES manual up              up
Serial0/1                 unassigned      YES unset  administratively down down
FastEthernet1/0          10.25.36.97     YES manual up              up
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 6 subnets, 5 masks
C       10.25.36.0/26 is directly connected, FastEthernet0/0
S       10.25.36.112/29 [1/0] via 10.25.36.122
S       10.25.36.124/30 [1/0] via 10.25.36.122
C       10.25.36.120/30 is directly connected, Serial0/0
C       10.25.36.96/28 is directly connected, FastEthernet1/0
C       10.25.36.64/27 is directly connected, FastEthernet0/1
R1#
```

R2

```
R2(config)#ip route 10.25.36.112 255.255.255.240 10.25.36.126
R2(config)#end
R2#write
*Mar 1 00:09:53.855: %SYS-5-CONFIG_I: Configured from console by console
R2#write memory
Building configuration...
[OK]
R2#show ip interface brief
Interface                IP-Address      OK? Method Status          Protocol
FastEthernet0/0          10.25.36.125    YES manual up              up
Serial0/0                 10.25.36.122    YES manual up              up
FastEthernet0/1          unassigned      YES unset  administratively down down
Serial0/1                 unassigned      YES unset  administratively down down
R2#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 6 subnets, 4 masks
S       10.25.36.0/26 [1/0] via 10.25.36.121
S       10.25.36.112/28 [1/0] via 10.25.36.126
C       10.25.36.124/30 is directly connected, FastEthernet0/0
C       10.25.36.120/30 is directly connected, Serial0/0
S       10.25.36.96/28 [1/0] via 10.25.36.121
S       10.25.36.64/27 [1/0] via 10.25.36.121
R2#
```

R3

```
R3
R1 R2 R3 PC1 PC2 PC3
R3(config)#ip route 10.25.36.112 255.255.255.248 10.25.36.125
R3(config)#end
R3#write
*Mar 1 00:12:58.691: %SYS-5-CONFIG_I: Configured from console by console
R3#write memory
Building configuration...
[OK]
R3#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 10.25.36.113    YES manual up          up
Serial10/0       unassigned      YES unset   administratively down down
FastEthernet0/1 10.25.36.126    YES manual up          up
Serial10/1       unassigned      YES unset   administratively down down
R3#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

 10.0.0.0/8 is variably subnetted, 6 subnets, 5 masks
S    10.25.36.0/26 [1/0] via 10.25.36.125
C    10.25.36.112/29 is directly connected, FastEthernet0/0
C    10.25.36.124/30 is directly connected, FastEthernet0/1
S    10.25.36.120/30 [1/0] via 10.25.36.125
S    10.25.36.96/28 [1/0] via 10.25.36.125
S    10.25.36.64/27 [1/0] via 10.25.36.125
R3#
```

PC1

```
PC1> ping 10.25.36.1
84 bytes from 10.25.36.1 icmp_seq=1 ttl=255 time=15.706 ms
84 bytes from 10.25.36.1 icmp_seq=2 ttl=255 time=16.113 ms
84 bytes from 10.25.36.1 icmp_seq=3 ttl=255 time=16.971 ms
84 bytes from 10.25.36.1 icmp_seq=4 ttl=255 time=17.625 ms
84 bytes from 10.25.36.1 icmp_seq=5 ttl=255 time=15.511 ms

PC1> ping 10.25.36.66
10.25.36.66 icmp_seq=1 timeout
84 bytes from 10.25.36.66 icmp_seq=2 ttl=63 time=32.110 ms
84 bytes from 10.25.36.66 icmp_seq=3 ttl=63 time=42.416 ms
84 bytes from 10.25.36.66 icmp_seq=4 ttl=63 time=27.730 ms
84 bytes from 10.25.36.66 icmp_seq=5 ttl=63 time=32.275 ms

PC1> ping 10.25.36.98
10.25.36.98 icmp_seq=1 timeout
10.25.36.98 icmp_seq=2 timeout
84 bytes from 10.25.36.98 icmp_seq=3 ttl=63 time=31.264 ms
84 bytes from 10.25.36.98 icmp_seq=4 ttl=63 time=31.565 ms
84 bytes from 10.25.36.98 icmp_seq=5 ttl=63 time=31.805 ms

PC1> ping 10.25.36.114
10.25.36.114 icmp_seq=1 timeout
10.25.36.114 icmp_seq=2 timeout
84 bytes from 10.25.36.114 icmp_seq=3 ttl=61 time=60.962 ms
84 bytes from 10.25.36.114 icmp_seq=4 ttl=61 time=64.767 ms
84 bytes from 10.25.36.114 icmp_seq=5 ttl=61 time=62.227 ms

PC1> trace 10.25.36.114
trace to 10.25.36.114, 8 hops max, press Ctrl+C to stop
 1  10.25.36.1    17.911 ms  9.380 ms  15.614 ms
 2  10.25.36.122  15.393 ms  15.570 ms  14.483 ms
 3  10.25.36.126  46.260 ms  45.847 ms  44.755 ms
 4  *10.25.36.114  60.418 ms (ICMP type:3, code:3, Destination port unreachable)

PC1> show ip

NAME       : PC1[1]
IP/MASK    : 10.25.36.2/26
GATEWAY    : 10.25.36.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT     : 10030
RHOST:PORT : 127.0.0.1:10031
MTU        : 1500

PC1> 
```

PC2

```

R1 R2 R3 PC1 PC2 PC3
PC2> show ip
NAME       : PC2[1]
IP/MASK    : 10.25.36.66/27
GATEWAY    : 10.25.36.65
DNS        :
MAC        : 00:50:79:66:68:01
LPORT      : 10028
RHOST:PORT : 127.0.0.1:10029
MTU        : 1500

PC2> ping 10.25.36.2
10.25.36.2 icmp_seq=1 timeout
10.25.36.2 icmp_seq=2 timeout
84 bytes from 10.25.36.2 icmp_seq=3 ttl=63 time=31.873 ms
84 bytes from 10.25.36.2 icmp_seq=4 ttl=63 time=32.752 ms
84 bytes from 10.25.36.2 icmp_seq=5 ttl=63 time=31.803 ms

PC2> trace 10.25.36.2
trace to 10.25.36.2, 8 hops max, press Ctrl+C to stop
 1  10.25.36.65  22.597 ms  7.661 ms  14.461 ms
 2  *10.25.36.2  32.099 ms (ICMP type:3, code:3, Destination port unreachable)

PC2>
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```

PC3

```

R1 R2 R3 PC1 PC2 PC3
PC3> show ip
NAME       : PC3[1]
IP/MASK    : 10.25.36.98/28
GATEWAY    : 10.25.36.97
DNS        :
MAC        : 00:50:79:66:68:02
LPORT      : 10032
RHOST:PORT : 127.0.0.1:10033
MTU        : 1500

PC3> ping 10.25.36.114
10.25.36.114 icmp_seq=1 timeout
10.25.36.114 icmp_seq=2 timeout
84 bytes from 10.25.36.114 icmp_seq=3 ttl=61 time=67.794 ms
84 bytes from 10.25.36.114 icmp_seq=4 ttl=61 time=64.567 ms
84 bytes from 10.25.36.114 icmp_seq=5 ttl=61 time=64.284 ms

PC3> trace 10.25.36.114
trace to 10.25.36.114, 8 hops max, press Ctrl+C to stop
 1  10.25.36.97  16.016 ms  13.985 ms  14.564 ms
 2  10.25.36.122 14.521 ms  14.215 ms  15.037 ms
 3  10.25.36.126 62.076 ms  47.044 ms  46.489 ms
 4  *10.25.36.114 76.785 ms (ICMP type:3, code:3, Destination port unreachable)

PC3>
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```

PC4

```
PC4> show ip

NAME       : PC4[1]
IP/MASK    : 10.25.36.114/30
GATEWAY    : 10.25.36.113
DNS        :
MAC        : 00:50:79:66:68:03
LPORT      : 10034
RHOST:PORT : 127.0.0.1:10035
MTU        : 1500

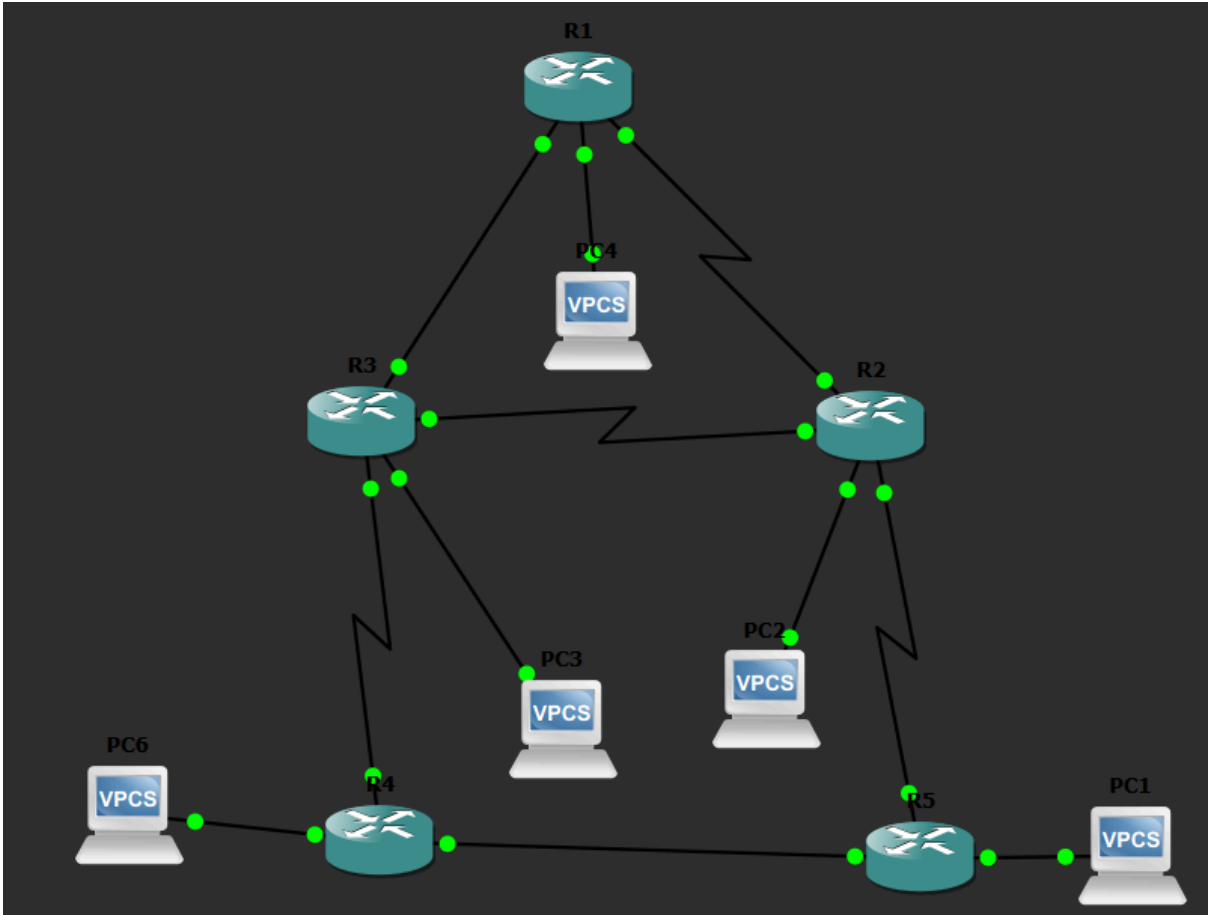
PC4> ping 10.25.36.2
10.25.36.2 icmp_seq=1 timeout
10.25.36.2 icmp_seq=2 timeout
84 bytes from 10.25.36.2 icmp_seq=3 ttl=61 time=63.960 ms
84 bytes from 10.25.36.2 icmp_seq=4 ttl=61 time=62.523 ms
84 bytes from 10.25.36.2 icmp_seq=5 ttl=61 time=62.543 ms

PC4> trace 10.25.36.2
trace to 10.25.36.2, 8 hops max, press Ctrl+C to stop
 1  10.25.36.113  16.511 ms  10.920 ms  14.236 ms
 2  10.25.36.125  45.732 ms  45.593 ms  46.320 ms
 3  10.25.36.121  46.304 ms  47.487 ms  46.003 ms
 4  *10.25.36.2  62.165 ms (ICMP type:3, code:3, Destination port unreachable)

PC4> 
```

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EJERICICIO 2



#2 Dinamico

~~10.25.36.0/24~~ 192.25.36.0/24

area	Red	mascara	primer host	ultimo host	broadcast	saltos	mask
lan PC1	192.25.36.0	255.255.255.192	.1	.62	.63	64 $n=6$	$32-6=26$
lan PC2	.64	.224	.65	.94	.95	32 $n=5$	$32-5=27$
lan PC3	.96	.240	.97	.110	.111	16 $n=4$	$32-4=28$
lan PC4	.112	.248	.113	.118	.119	8 $n=3$	$32-3=29$
lan PC6	.120	.248	.121	.126	.127	8 $n=3$	29
R1-R3	.128	.252	.129	.130	.131	4 $n=2$	30
R1-R2	.132	.252	.133	.134	.135	4	
R3-R2	.136	.252	.137	.138	.139	4	
R3-R4	.140	.252	.141	.142	.143	4	
R4-R5	.144	.252	.145	.146	.147	4	
R5-R2	.148	.252	.149	.150	.151	4	

tabla IP	inter	IP	mask	gateway
lan PC1 R5	Fa0/0	.1	.192	—
PC1	NIC	.2		.1
lan PC2 R2	Fa0/0	.65	.224	—
PC2	NIC	.66		.65
lan PC3 R3	Fa0/0	.97	.240	—
PC3	NIC	.98		.97
lan PC4 R1	Fa0/0	.113	.248	—
PC4	NIC	.114		.113
lan PC6 R4	Fa0/0	.121	.248	—
PC6	NIC	.122		.121
R1-R3 R1	Fa0/1	.129	.252	—
R3	Fa0/1	.130		—
R3-R2 R3	so/0	.137	"	—
R2	so/0	.138		—
R3-R4 R3	so/1	.141	"	—
R4	so/0	.142		—
R4-R5 R4	Fa0/1	.145	"	—
R5	Fa0/1	.146		—
R5-R2 R5	so/0	.149	"	—
R2	so/2	.150		—
R1-R2 R1	so/0	.133	"	—
R2	so/1	.134		—